

JUNO Reactor Antineutrino + Neutron Monte Carlo

Progressive simulation report

Detector mass: 20.0 kt

Generated MC events: 100,000

Selected IBD-like events: 99,002

Selection efficiency: 0.9900

Estimated free protons: $1.445\text{e}+33$

Oscillation hierarchy used for MC: NO

Note: phenomenological reactor, neutron-transport and detector-response model.

Simulation Configuration

Simulation configuration

Scintillator mass = 20.00 kt

Target radius = 17.70 m

Energy resolution coefficient = 0.0302

Light yield = 1345.0 PE/MeV

Hydrogen-target approximation = 12% mass fraction

Neutron capture time constant = 200.0 μ s

Neutron capture gamma energy = 2.223 MeV

Reactor configuration

Yangjiang: 17.40 GWth, 52.80 km

Taishan: 9.20 GWth, 53.10 km

Fission fractions

U235: 0.550

U238: 0.080

Pu239: 0.300

Pu241: 0.070

IBD selection

Prompt: 0.70-8.00 MeV

Delayed: 1.80-2.60 MeV

Time window: 1000.0 μ s

Spatial cut: 2.00 m

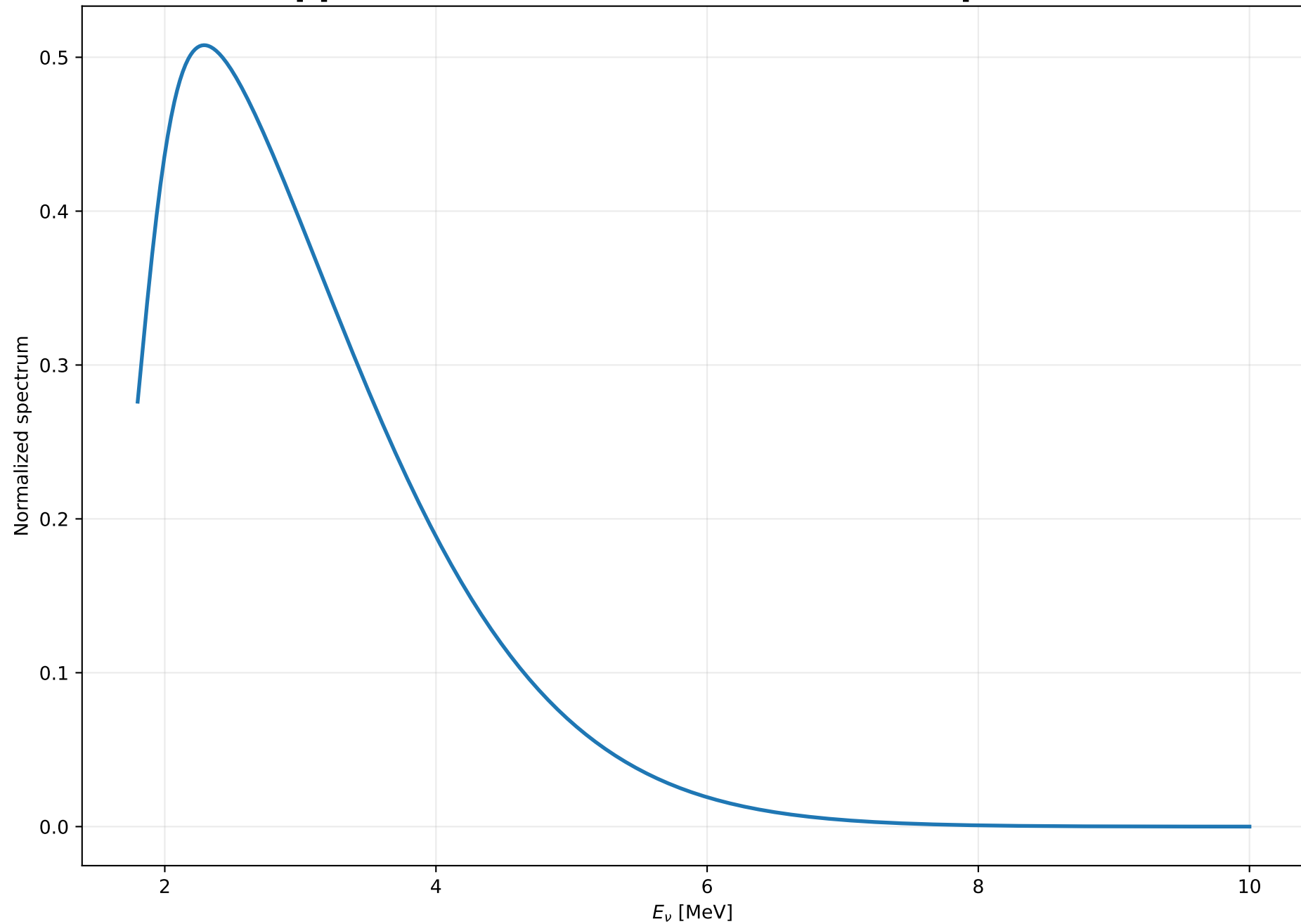
Monte Carlo Result Summary

Quantity	Value
Generated events	100000
Selected events	99002
Selection efficiency	0.99002
Mean neutrino energy [MeV]	4.65125
Mean prompt energy [MeV]	3.86893
Mean delayed energy [MeV]	2.22298
Mean neutron initial energy [keV]	53.1046
Mean capture time [us]	199.03
Mean capture displacement [cm]	0.282595
Estimated free protons	1.44531e+33

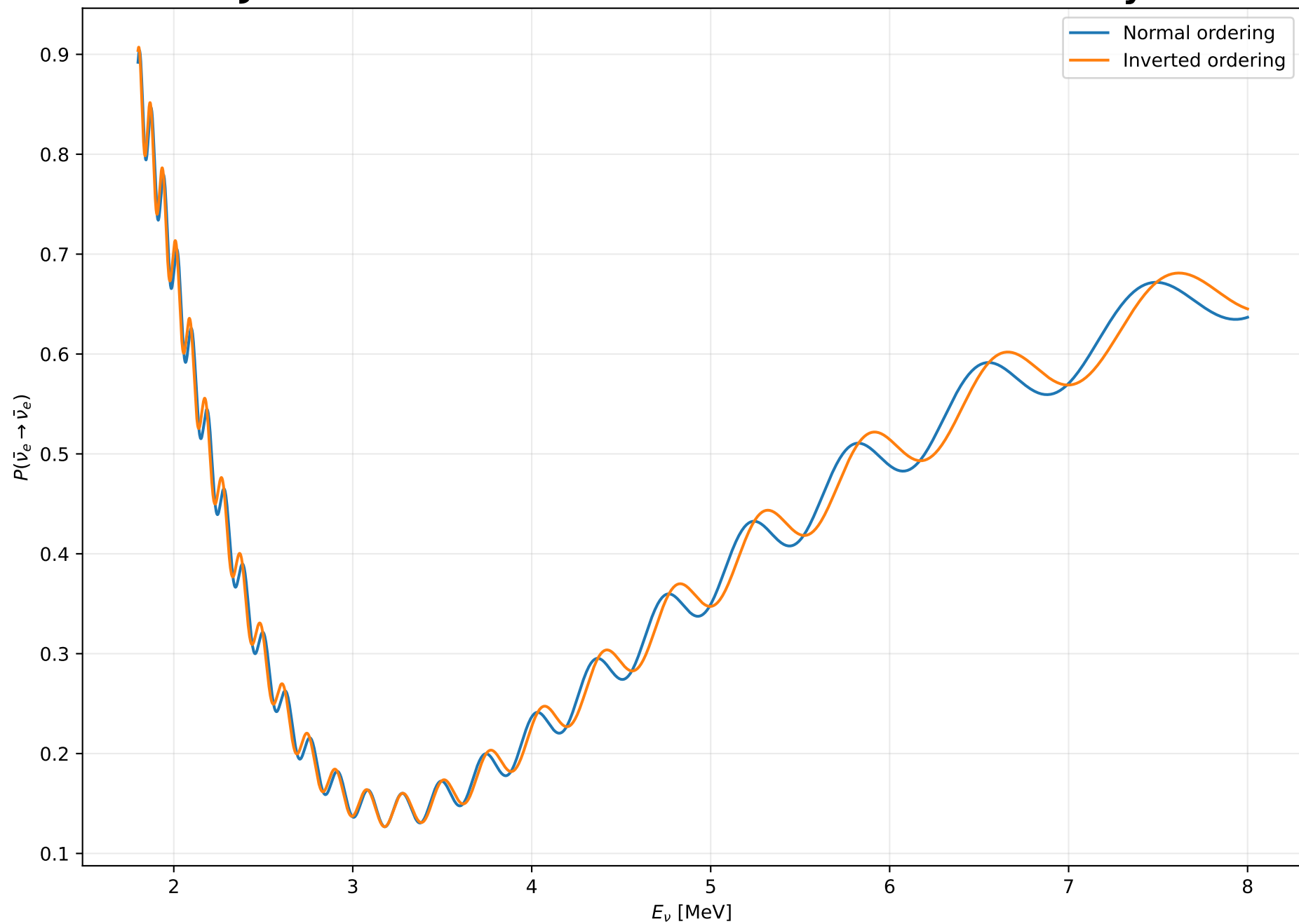
Sample of Generated Events

E_nu_MeV	E_prompt_reco_MeV	E_delayed_reco_MeV	neutron_initial_MeV	capture_time_us	capture_distance_m
5.703555	4.927138	2.16803	0.149734	388.455881	0.003984
4.42803	3.638662	2.254275	0.084371	9.039207	0.000787
6.173114	5.46171	2.242379	0.095271	45.789666	0.001446
5.359837	4.633457	2.232714	0.060179	132.443586	0.001319
2.611762	1.869888	2.249814	0.016471	57.958044	0.002646
7.560569	6.866914	2.266914	0.087394	723.094135	0.006229
5.645315	4.890706	2.191078	0.077444	399.046199	0.003364
5.75929	4.95539	2.216357	0.089109	13.266176	0.001005
2.847814	2.07658	2.32342	0.030837	375.33056	0.004185
4.470008	3.649071	2.208922	1e-05	57.280374	0.001613
4.184411	3.374721	2.277323	0.067481	107.587099	0.003421
6.734634	5.965056	2.190335	0.055043	40.889435	0.001101

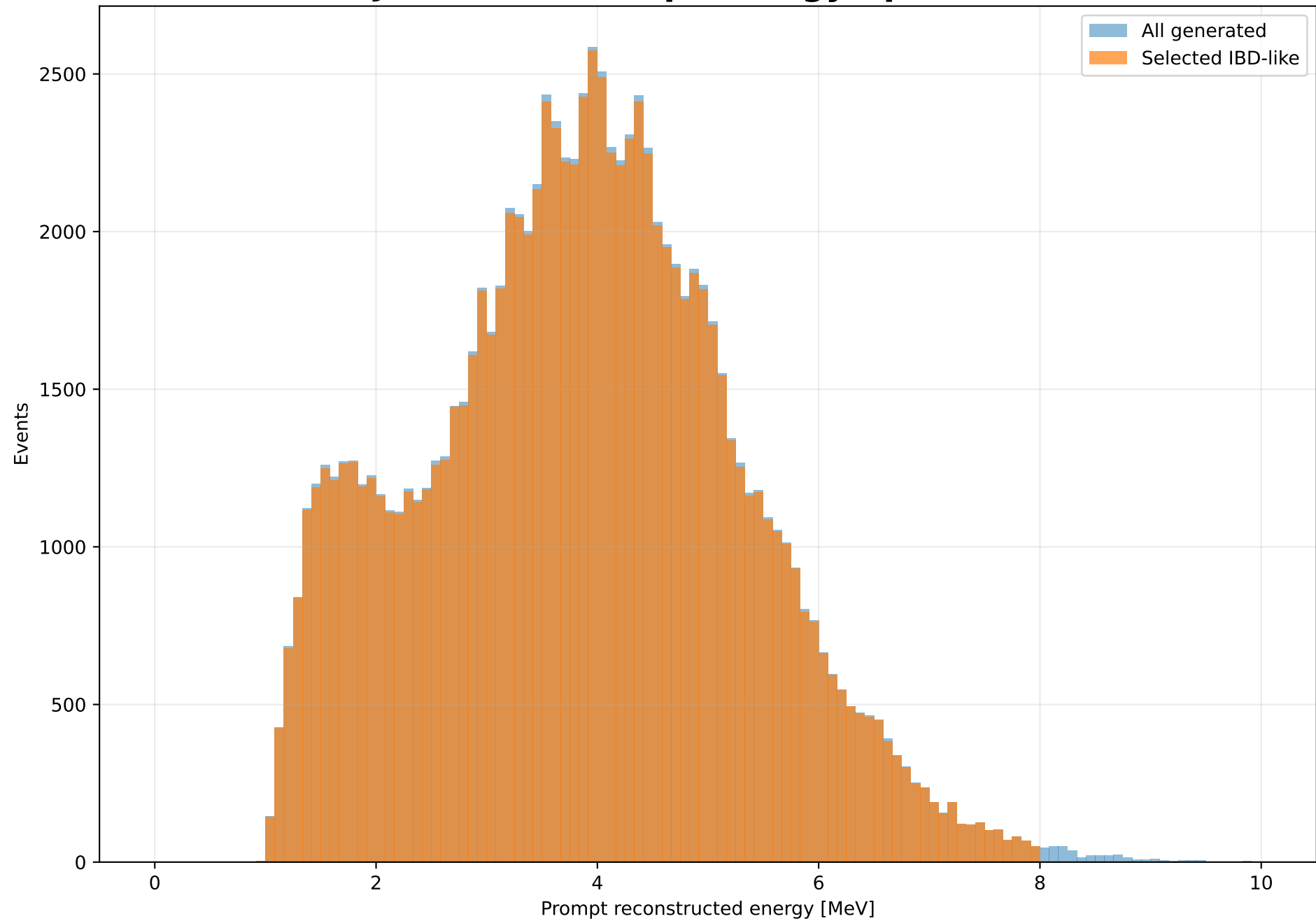
Approximate Reactor Antineutrino Spectrum



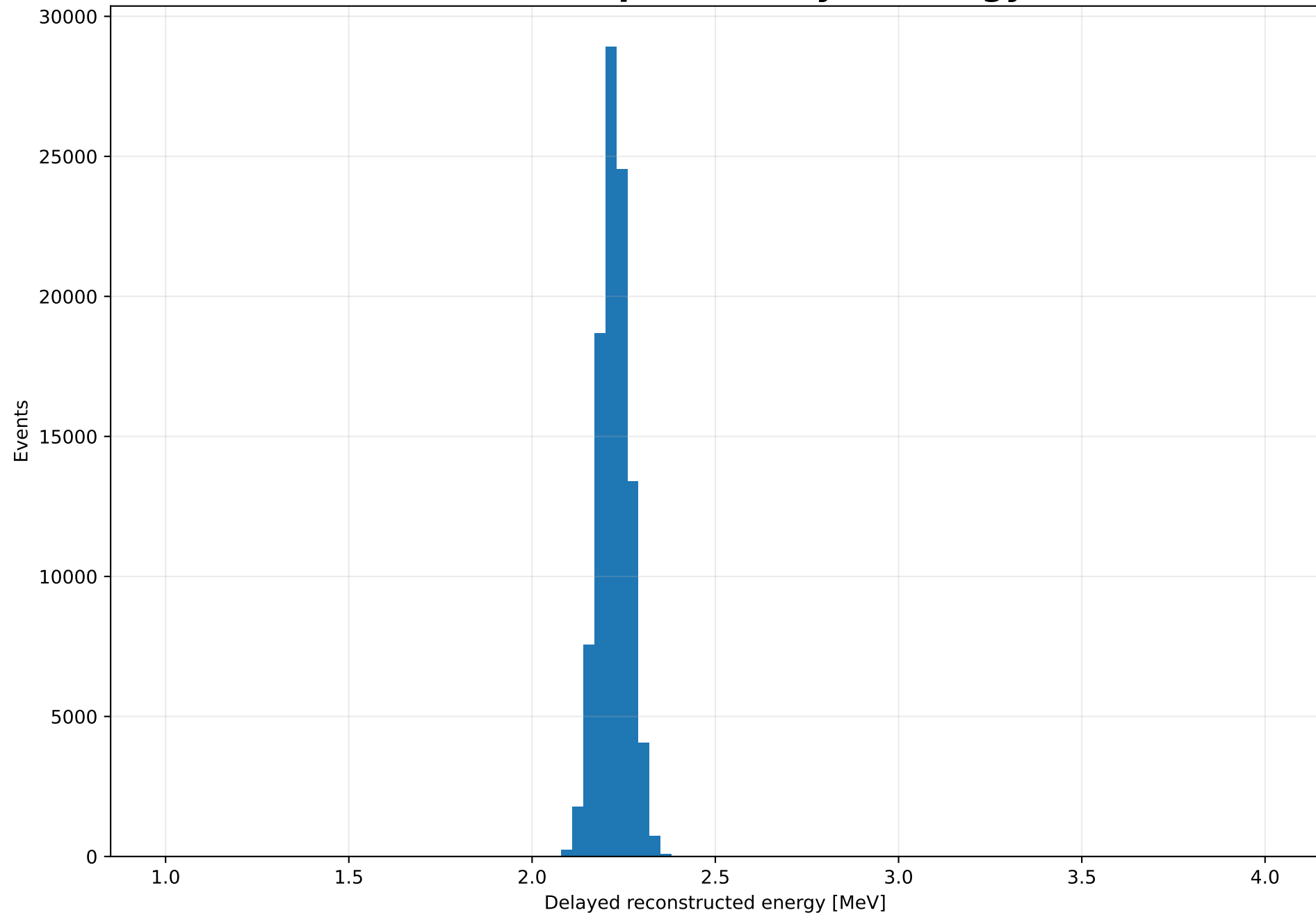
JUNO Electron Antineutrino Survival Probability



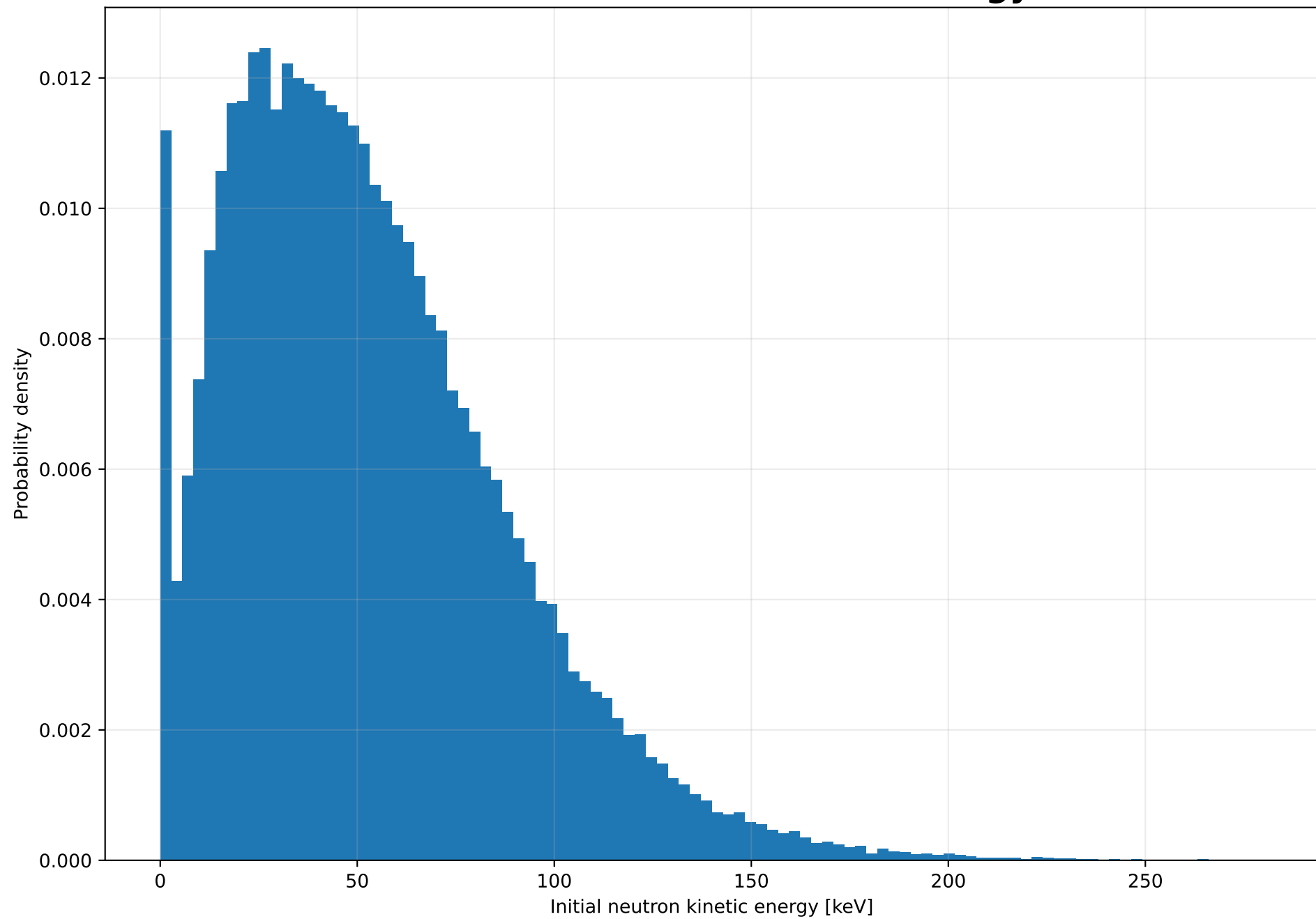
JUNO-like Prompt Energy Spectrum



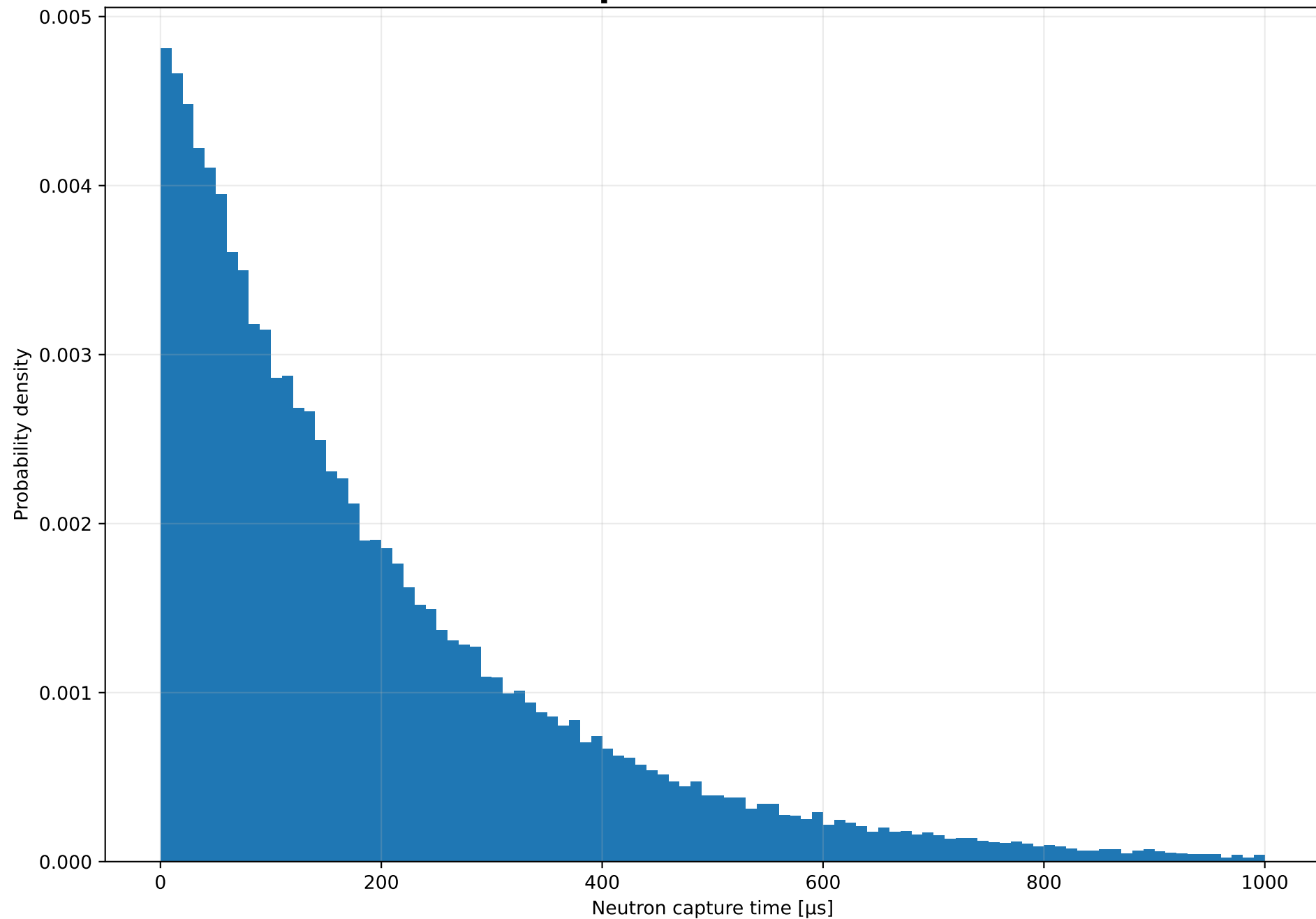
Neutron-Capture Delayed Energy



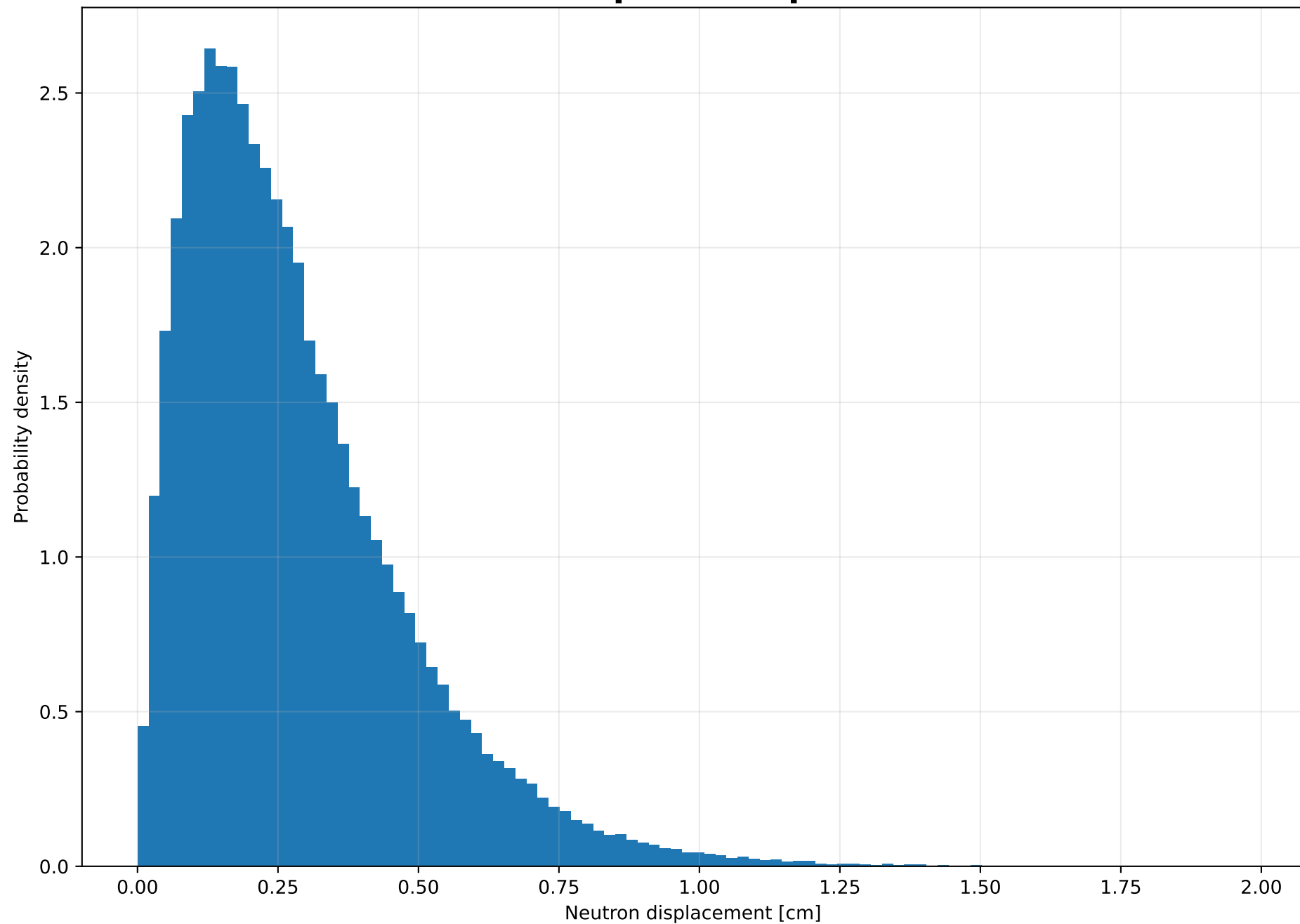
IBD Neutron Initial Kinetic Energy



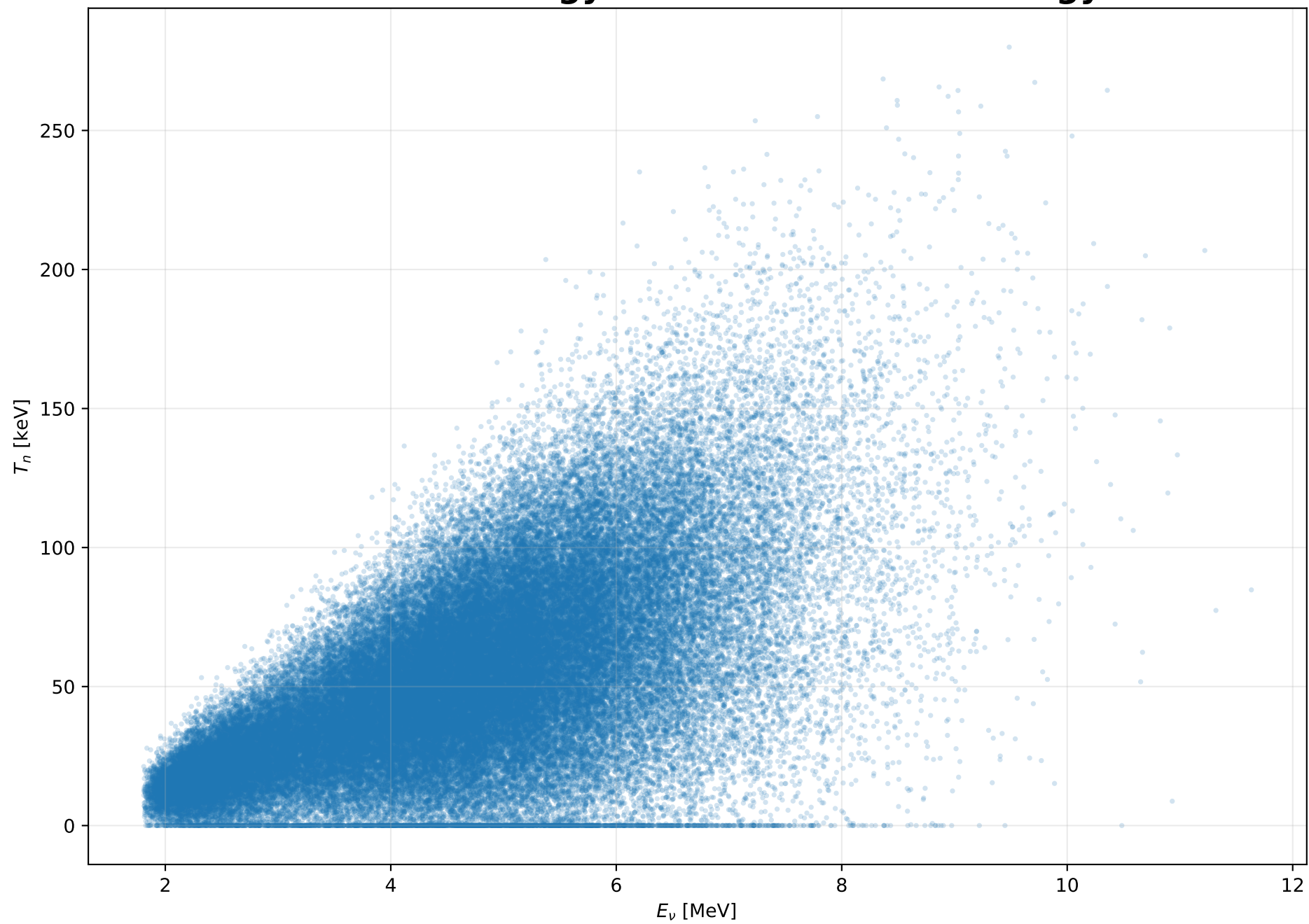
Neutron Capture-Time Distribution



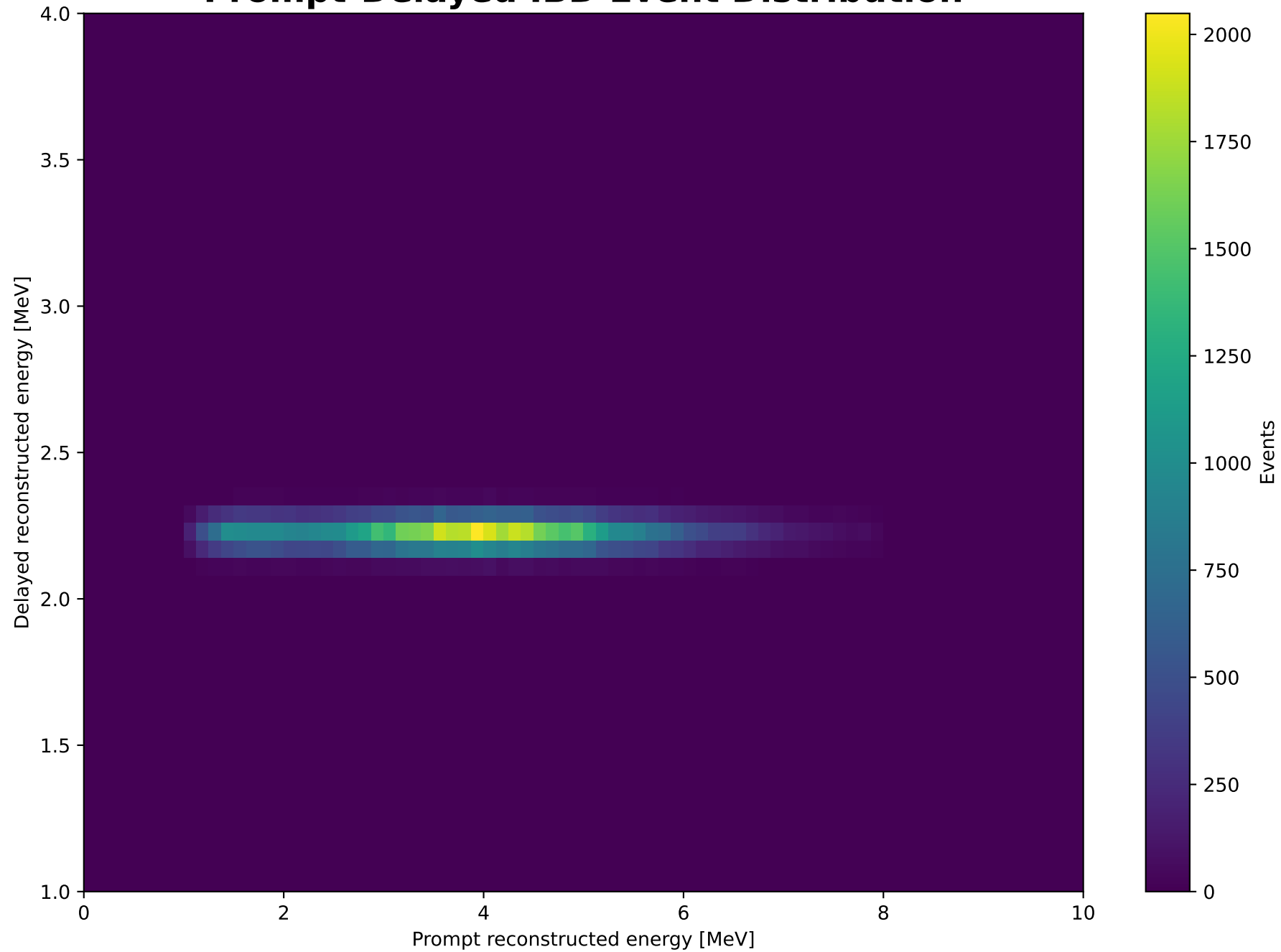
Neutron Capture Displacement



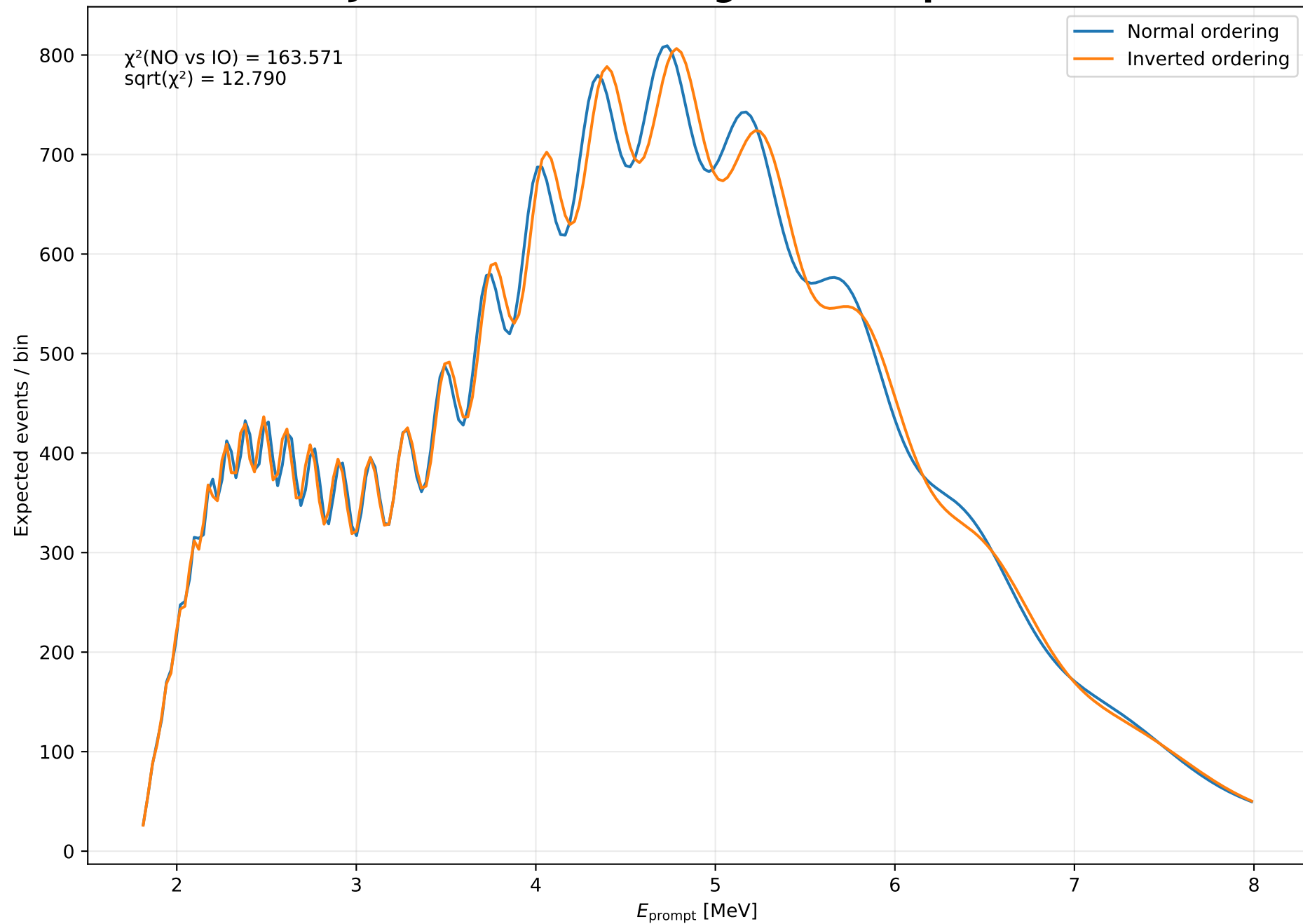
Neutrino Energy vs. Initial Neutron Energy



Prompt-Delayed IBD Event Distribution



JUNO Mass-Ordering Asimov Spectra



Interpretation and Limitations

The simulation is designed as a progressive JUNO/neutrino-physics study.

The reactor spectrum is phenomenological and not a Huber-Mueller or official JUNO flux prediction.

The IBD cross section is a simplified leading approximation.

The neutron moderation model is phenomenological and is NOT Geant4 neutron transport. It does not reproduce detailed material-dependent elastic-scattering and capture cross sections.

The detector response uses a simplified JUNO-like photoelectron yield and energy-resolution model.

Mass-ordering discrimination here is an idealized Asimov toy study.

Real JUNO sensitivity requires systematics, nonlinearity, reactor model uncertainties, backgrounds, detector response, and proper fitting.